

AgriSuit

by



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AGENDA



**Problem
Statement**

Motivation

Methodology

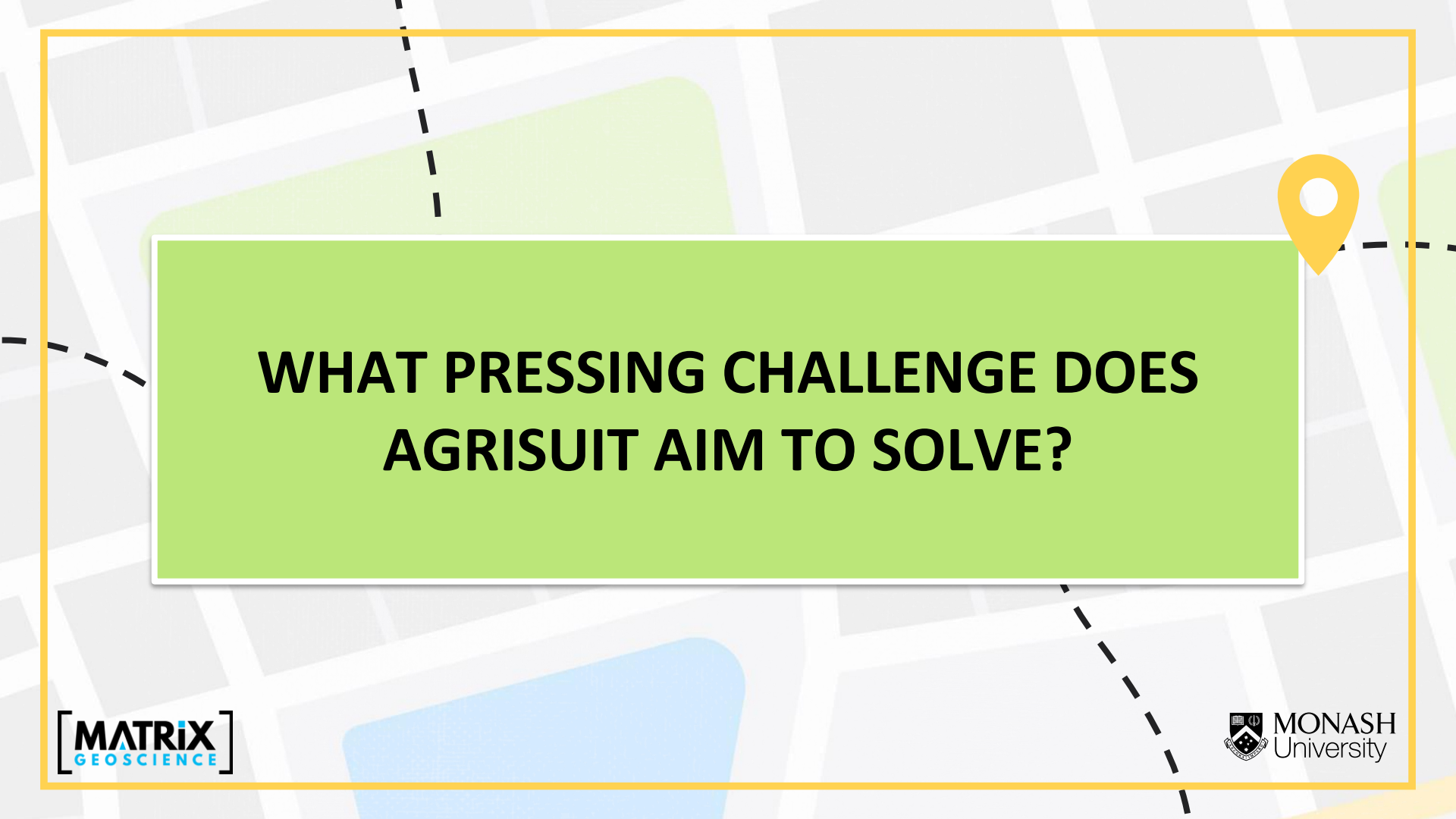
**Limitations &
Demonstration**

**Future Scope &
Conclusion**

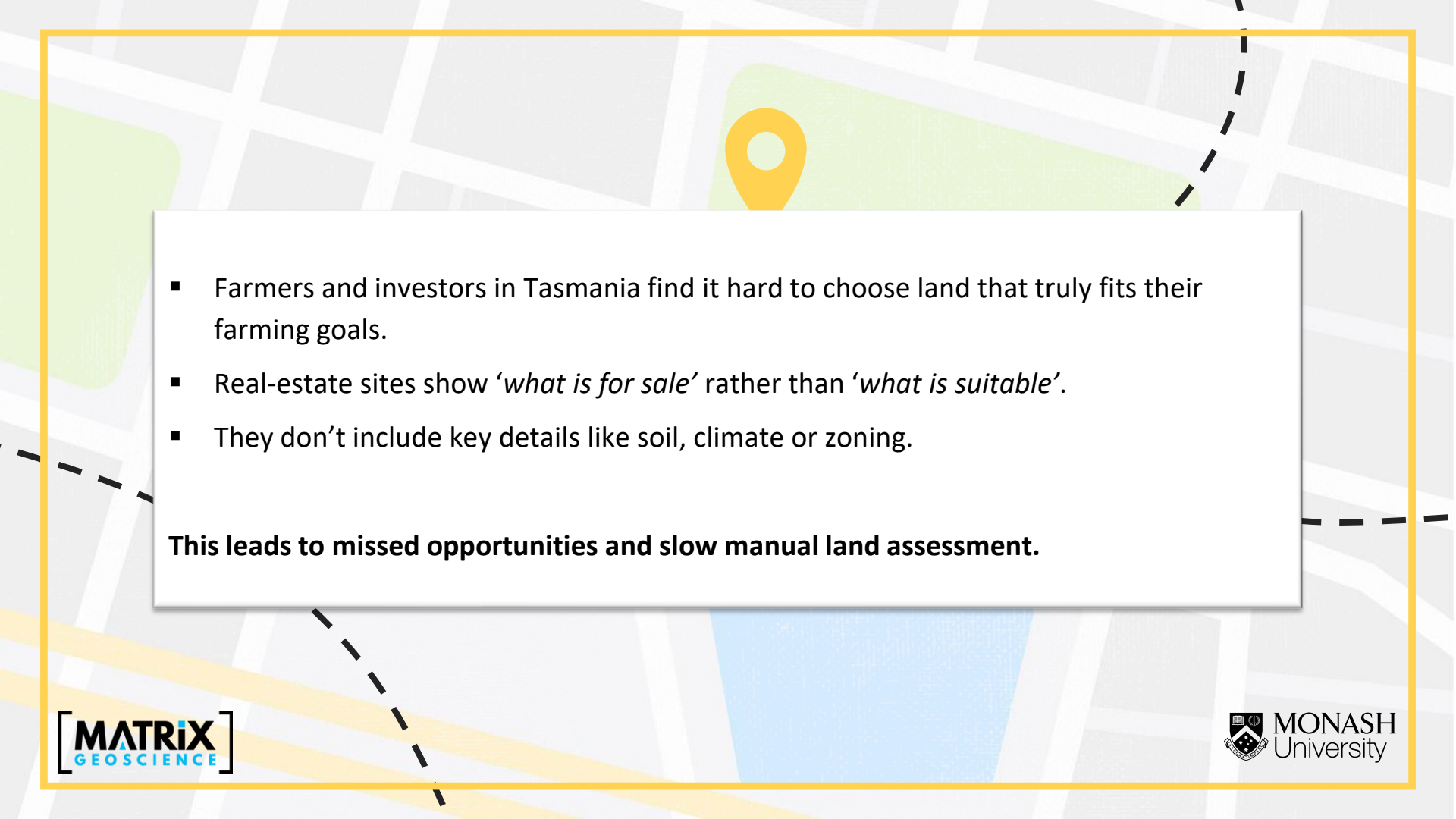
MATRIX GEOSCIENCE

Matrix Geoscience is an Australian consultancy delivering practical, mathematical solutions to streamline operations.

- Serves mining, geology, education, materials handling globally.
- Geo-statistics, data analytics, and ML expertise.
- Tailored, efficiency-focused implementations.

The background is a stylized map with various colored regions (green, blue, grey) and dashed lines. A large green rectangle is centered on the slide, containing the main text. An orange location pin icon is positioned in the upper right corner of the green rectangle.

WHAT PRESSING CHALLENGE DOES AGRISUIT AIM TO SOLVE?

- 
- Farmers and investors in Tasmania find it hard to choose land that truly fits their farming goals.
 - Real-estate sites show '*what is for sale*' rather than '*what is suitable*'.
 - They don't include key details like soil, climate or zoning.

This leads to missed opportunities and slow manual land assessment.

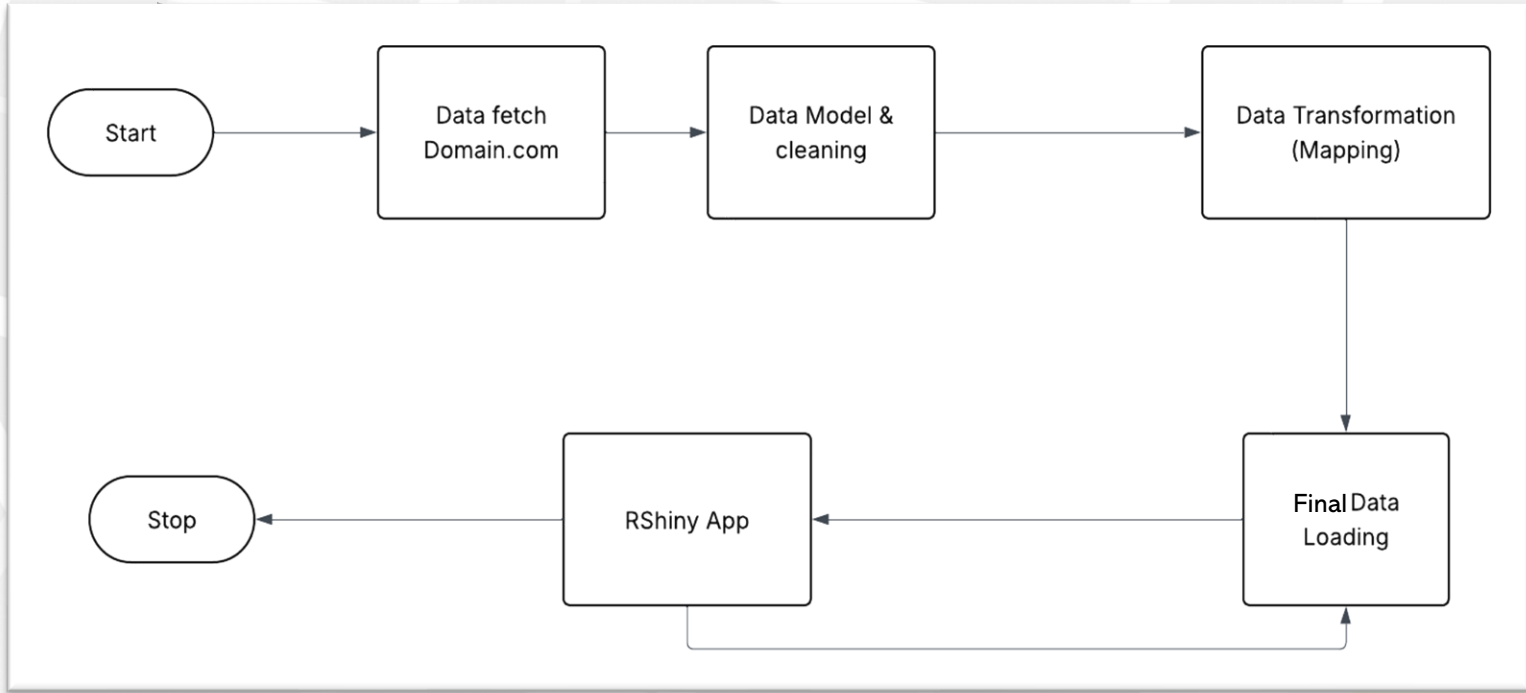
MOTIVATION

- Agriculture contributes ~7–9% to Tasmania's GDP, driving regional growth (Department of Treasury and Finance Tasmania, 2024)
- TESM (Tasmanian Enterprise Suitability Map) not linked to live property markets.
- AgriSuit bridges that gap, combining TESM data with property listings.
- Enables data-driven and climate-resilient land investment.

The background of the slide is a stylized map with a grid of light gray streets. A dashed black line winds across the map, starting from the left, curving around a yellow location pin in the upper left, and then continuing towards the bottom right. A solid orange line forms a rectangular border around the entire slide content.

METHODOLOGY

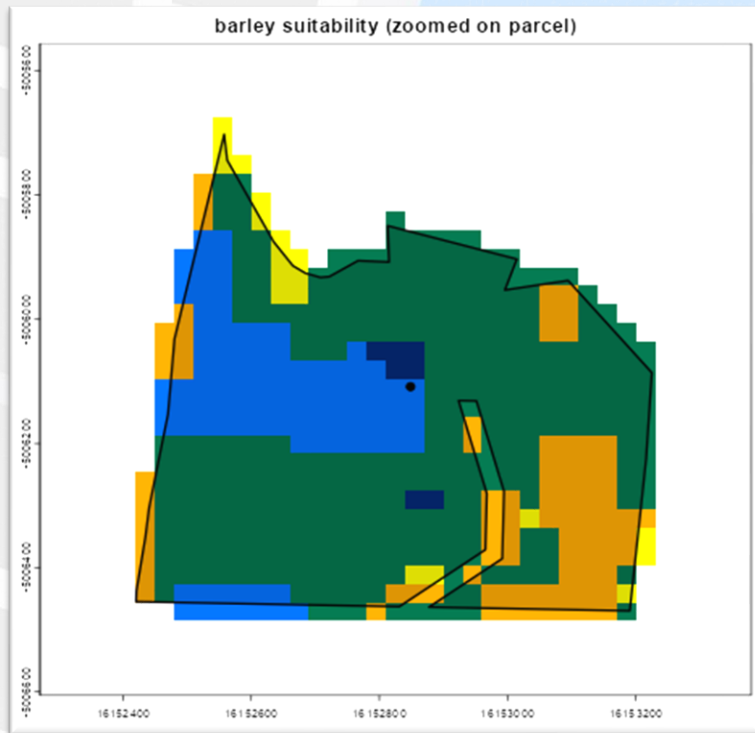
PROJECT FLOW



DATA SOURCES

- **Domain.com.au** for *property listings data*.
- **LISTMap open data** in for the *Statewide Cadastral Parcel, Address Points, and Crop Rules / Suitability Map*.
- Basemaps (Web-hosted Maps): **CartoDB Positron; LIST Topographic, LIST Orthophoto**.

PIXEL MAPPING



ESM suitability classes

- 1.0 Well suited
- 1.1 Well suited (with soil management)
- 2.0 Suitable
- 2.1 Suitable (with soil management)
- 3.0 Moderately suitable
- 3.1 Moderately suitable (with soil management)
- 4.0 Unsuitable

FINAL TABLE

FULL_ADDRESS	MUNICIPALITY	CROP	CLASS_LABEL	PRICE (AUD)	COUNT	COVERAGE (%)	WEBSITE
120 Russells Plains Road Rocherlea TAS 7248 Australia	LAUNCESTON	Barley	1.1 Well suited (with soil management)	550,000	31	8.2	View
120 Russells Plains Road Rocherlea TAS 7248 Australia	LAUNCESTON	Barley	4.0 Unsuitable	550,000	239	63.4	View
120 Russells Plains Road Rocherlea TAS 7248 Australia	LAUNCESTON	Barley	2.0 Suitable	550,000	105	28	View
167 Adams Road Lebrina TAS 7254 Australia	LAUNCESTON	Barley	1.1 Well suited (with soil management)	3,000,000	20	6.3	View
167 Adams Road Lebrina TAS 7254 Australia	LAUNCESTON	Barley	2.1 Suitable (with soil management)	3,000,000	62	19.6	View

WEB MAP SERVICE MAPPING



CartoDB Map



LIST Satellite



LIST Topography

LIMITATIONS

- 
- Climate projection only uses the RCP 8.5 climate pathway for 2018.
 - Absence of Yield or Productivity Estimates.
 - Deterministic Threshold Rules (No Uncertainty Propagation).
 - Crop options are limited to 33 crops on the current TESM model.



DEMONSTRATION

FUTURE SCOPE

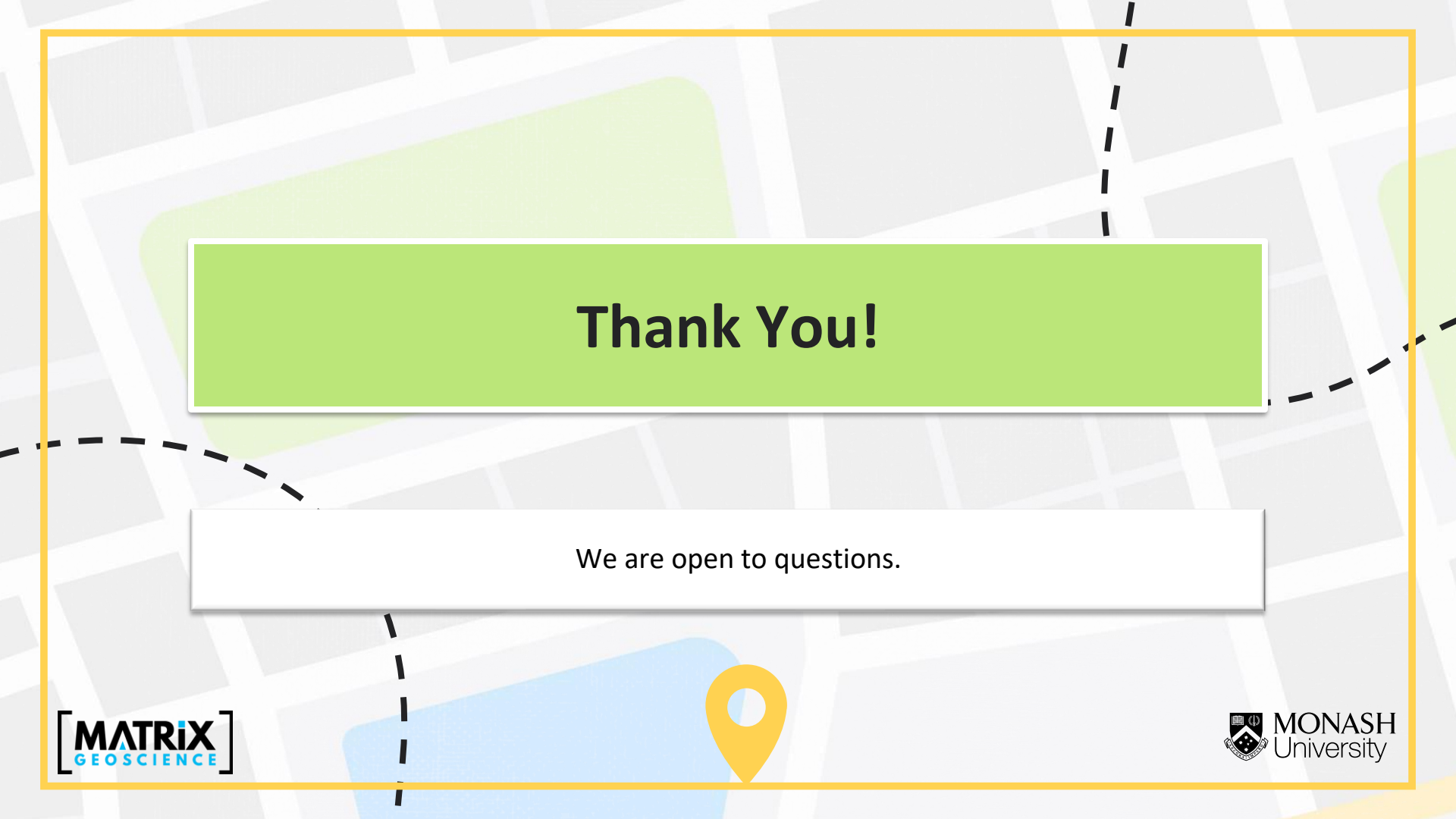
- Add 2030 & 2050 climate projections.
- Enable auto database refresh and email alerts.
- Expand to mainland Australia.
- Personalised E-mail based Notification

COMMERCIALISATION

- Replace scraping with API integrations for agricultural land listings.
- Move to cloud-based backend for scalability.
- Add premium features (ROI analysis, alerts, crop comparison).
- Transition toward a fully commercial, investor-ready platform.

CONCLUSION

- AgriSuit bridges the gap between geospatial intelligence and the property market.
- Empowers investors and farmers with science-backed land insights.
- Demonstrates how open data can drive sustainable growth.
- “AgriSuit. Where Every Land Finds Its Purpose.”

The background is a stylized map with grey streets, green parks, and blue water. A large green rectangle is centered in the upper half, and a yellow location pin is in the lower center. Dashed black lines are scattered across the map.

Thank You!

We are open to questions.